Project: Worked examples
Title: Lifting lug and pin joint
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A pinned lug between two cheek plates under a load along the lug: tension on the net section through the hole, bearing of the pin on the hole, shear tear-out beyond the hole, and the pin in double shear and in bending, each against an allowable stress; the largest utilisation governs.

WHAT IT CALCULATES	RELATION
Tension across the net section beside the hole	$\sigma_t = P/((W - d_h)\cdot t)$
Bearing of the pin on the lug	$\sigma_{br} = P/(d_p\cdot t)$
Tear-out shear beyond the hole, on two planes	$\tau_{to} = P/(2\cdot a\cdot t)$
Pin shear, in double shear	$\tau_p = P/(2\cdot\pi\cdot d_p^2/4)$
Pin bending, from the lever arm across the lug, the gap and the shackle	$M = (P/2)\cdot(t/4 + g + t_s/2)$; $\sigma_b = 32\cdot M/(\pi\cdot d_p^3)$
Utilisation of each mode, its stress over the allowable from the material and design category in force (the sheet adds no design factor of its own)	$u = \sigma/\sigma_{allow}$

Basis: Pinned connections[1](Ch. 3); Shear, bearing and bending stresses[2](Ch. 3).

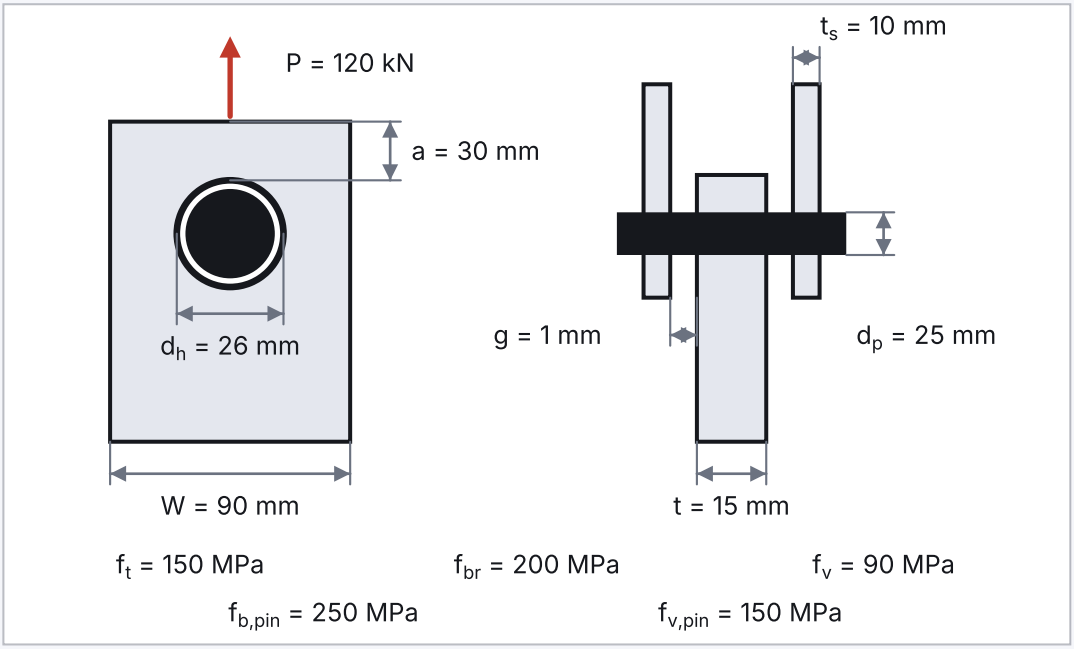


Figure 1 — A lug and its pin, front view and section through the pin: the lug between two cheek plates, the load through the pin. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P	Load on the lug (along the lug)	120	kN	spec: LOAD-SCHEDULE-01	0.01 … 100000 kN
t	Lug thickness	15	mm	drawing: DRG-LL-102 rev A	1 … 500 mm
W	Lug width at the hole	90	mm	drawing: DRG-LL-102 rev A	5 … 2000 mm
d_h	Hole diameter	26	mm	drawing: DRG-LL-102 rev A	2 … 1000 mm
a	Edge distance, hole edge to lug end (load direction)	30	mm	drawing: DRG-LL-102 rev A	1 … 2000 mm
d_p	Pin diameter	25	mm	drawing: DRG-LL-102 rev A	2 … 1000 mm
t_s	Cheek plate thickness (each)	10	mm	drawing: DRG-LL-102 rev A	1 … 500 mm
g	Clearance, lug face to cheek plate (each side)	1	mm	drawing: DRG-LL-102 rev A	0 … 100 mm
f_t	Allowable tensile stress, lug	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
f_{br}	Allowable bearing stress, lug	200	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa
f_v	Allowable shear stress, lug	90	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
$f_{b,pin}$	Allowable bending stress, pin	250	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa
$f_{v,pin}$	Allowable shear stress, pin	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa

3. Assumptions

A1. The load acts along the lug's axis through the hole centre. (No out-of-plane or side load.)
A2. The pin bears uniformly over the lug thickness and over each cheek plate; each cheek plate reacts half the load. (Pin-bending idealisation for a lug between two cheek plates.)
A3. A close-fitting pin ($d_p \geq 0.9 \cdot d_h$). (Uniform-bearing idealisation.)

4. Working

[14] $k_{fit} = d_p / d_h$
$= 25\ mm / 26\ mm$
$= 0.96154$
Pin-to-hole ratio

[15] $A_{net} = (W - d_h) * t$
$= (90\ mm - 26\ mm) \times 15\ mm$
$= 960\ mm^2$
Net section through the hole

[16] $\sigma_t = P / A_{net}$ [1](Ch. 3)
$= 120\ kN / 960\ mm^2$
$= 125\ MPa$
Net-section tension

[17] $A_{br} = d_p * t$
$= 25\ mm \times 15\ mm$
$= 375\ mm^2$
Bearing area

[18] $\sigma_{br} = P / A_{br}$
$= 120\ kN / 375\ mm^2$
$= 320\ MPa$
Bearing of the pin on the hole

[19] $A_{to} = 2 * a * t$
$= 2 \times 30\ mm \times 15\ mm$
$= 900\ mm^2$
Tear-out area (two planes beyond the hole)

[20] $\tau_{to} = P / A_{to}$
$= 120\ kN / 900\ mm^2$
$= 133.33\ MPa$
Tear-out shear

[21] $A_p = 2 * pi * d_p^2 / 4$
$= 2 \times \pi \times (25\ mm)^2 / 4$
$= 981.75\ mm^2$
Pin area in double shear

[22] $\tau_p = P / A_p$
$= 120\ kN / 981.75\ mm^2$
$= 122.23\ MPa$
Pin shear

[23] $M_p = P / 2 * (t / 4 + g + t_s / 2)$
$= 120\ kN / 2 \times (15\ mm / 4 + 1\ mm + 10\ mm / 2)$
$= 585\ N·m$
Pin bending moment (uniform bearing on lug and cheek plates)

[24] $Z_p = pi * d_p^3 / 32$
$= \pi \times (25\ mm)^3 / 32$
$= 1534\ mm^3$
Pin section modulus

[25] $\sigma_b = M_p / Z_p$
$= 585\ N·m / 1534\ mm^3$
$= 381.36\ MPa$
Pin bending stress

[26] $u_t = \sigma_t / f_t$
$= 125\ MPa / 150\ MPa$
$= 0.83333$
Net-section utilisation

[27] $u_{br} = \sigma_{br} / f_{br}$
$= 320\ MPa / 200\ MPa$
$= 1.6$
Bearing utilisation

[28] $u_{to} = \tau_{to} / f_v$
= 133.33 MPa / 90 MPa
= 1.4815
Tear-out utilisation
[29] $u_p = \tau_p / f_{v,pin}$
= 122.23 MPa / 150 MPa
= 0.81487
Pin shear utilisation
[30] $u_b = \sigma_b / f_{b,pin}$
= 381.36 MPa / 250 MPa
= 1.5254
Pin bending utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
σ_t	125 MPa	Net-section tension
σ_{br}	320 MPa	Bearing stress
τ_{to}	133.33 MPa	Tear-out shear
τ_p	122.23 MPa	Pin shear
M_p	585 N·m	Pin bending moment
σ_b	381.36 MPa	Pin bending stress
u_t	0.83333	Net-section utilisation
u_{br}	1.6	Bearing utilisation
u_{to}	1.4815	Tear-out utilisation
u_p	0.81487	Pin shear utilisation
u_b	1.5254	Pin bending utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Bearing	u_{br} = 1.6	$u_{br} \le 1$	1.6	[1](Ch. 3)
FAIL	Pin bending	u_b = 1.5254	$u_b \le 1$	1.53	
FAIL	Tear-out	u_{to} = 1.4815	$u_{to} \le 1$	1.48	[1](Ch. 3)
PASS	Net-section tension	u_t = 0.83333	$u_t \le 1$	0.833	[1](Ch. 3)
PASS	Pin double shear	u_p = 0.81487	$u_p \le 1$	0.815	

7. Applicability

No warnings.

8. Verification

Template mech.lifting-lug@1.0.1 Content hash 17ed215aa272cf35 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. American Society of Mechanical Engineers. Design of Below-the-Hook Lifting Devices. ASME BTH-1. ASME.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.